Contents

Part I Differentiation: Mathematical Description of Motion

1 Limit and Continuity #1 (with Common Language) 3
1.1 Common Language Definitions 4
1.2 ε–δ arguments for quality control 9

2 Limit and Continuity #2 (with ε–δ argument) 15
2.1 Rigorous definitions using ε–δ 15
2.2 Examples 20
2.3 Limits as $x \to \infty$ and $f(x) \to \infty$ 22

3 Differentiation 25
3.1 Rate of Increase 25
3.2 Differentiation Rules 28
3.3 Intermediate and Mean Value Theorem 32

4 Chain Rule and Variable-Centered Notation 37
4.1 Chain Rule 37
4.2 Variable-Centered Notation 42

5 Integration & Fundamental Theorem of Calculus 49
5.1 Antiderivative 49
5.2 Area Function as an Integral 51
5.3 Riemann Sum and Area 53

6 Inverse Functions and Their Derivatives 63
6.1 Graph and Differentiation 63
6.2 Inverse Functions in Variable-Centered Notation 67
6.3 Integration by change of variables 68

7 Logarithm, Exponential, & Implicit Differentiation 73
7.1 Natural logarithm 73
7.2 Exponential function 76

7.3 Implicit differentiation 78

Part II Kepler and Newton's Laws of Motion

8 Rectangular Coordinate System and Curves in $\mathbb{R}^3$ 85
8.1 Projection and coordinate system 85
8.2 Vector Space 90
8.3 Inner Product 91
8.4 Cross product 94

9 Polar coordinates in $\mathbb{R}^2$ 99
9.1 Moving particle and trajectory curves in space 99
9.2 Polar coordinates 100
9.3 Motion in polar coordinates 103
9.4 Ellipse in polar coordinates 105
9.5 Curves in polar coordinates (optional) 107

10 First Order Differential Equations 111
10.1 First order differential equations 111
10.2 Separation of variables 116
10.3 First-Order Linear Equation and Integrating Factor 118

11 Second Order Differential Equations 123
11.1 Second Order Linear Equation 123
11.2 Homogeneous Problem and Characteristic Polynomial 127
11.3 Initial value problem 131

12 Newton's Law on Earth's Surface 135
12.1 Newton's law of motion and gravitation 135
12.2 Work and energy 136
12.3 Gravity force and potential energy on Earth 137
12.4 Projectile motion on Earth 139

13 Newton's Law in Space: Two-Body Problem 145
13.1 Two-body problem 145
13.2 Center of mass (barycenter) 146
13.3 Kepler problem 148
13.4 Orbit of ICBM #1 (Optional) 152

14 Kepler's Law (Optional) 157
14.1 Kepler's first law and elliptical orbits 157
14.2 Examples and Applications 161

Part III The Arts of Calculus

15 Curves and Particle Trajectories in $\mathbb{R}^3$ 169
15.1 Arclength as a variable 169
15.2 TNB coordinate system 172

15.3 Computation formulas (optional) . . . 176

16 Linearization and Differentiation . . . 181
16.1 Linearization . . . 181
16.2 Differentials . . . 183
16.3 Differentials for linear approximation . . . 185

17 Inverse trigonometric and hyperbolic functions . . . 189
17.1 Inverse trigonometric functions . . . 189
17.2 Hyperbolic functions . . . 194

18 L'Hôpital's rule and big-oh / little-oh . . . 199
18.1 L'Hôpital's rule . . . 199
18.2 Calculating limits using inverse functions . . . 203
18.3 Big-oh and little-oh . . . 204

19 Integration by substitution and by parts . . . 207
19.1 Substitution . . . 208
19.2 Integration by parts . . . 210
19.3 Trigonometric substitution . . . 213

20 Rational Functions and Improper Integration . . . 217
20.1 Integration of rational functions . . . 217
20.2 Integral over unbounded domains . . . 221
20.3 Integral of unbounded functions . . . 224

Part IV Approximation Techniques and Series

21 Numerical Integration . . . 231
21.1 Numerical integration and Riemann sum . . . 231
21.2 Convergence order . . . 233
21.3 Numerical integration and Gauss–Legendre quadrature . . . 236

22 Sequences and Series . . . 243
22.1 Sequence of real numbers . . . 243
22.2 Series of real numbers . . . 248
22.3 Power series . . . 250

23 Tests for Absolute Convergence . . . 255
23.1 Integral Test . . . 255
23.2 Comparison Test . . . 257
23.3 Ratio test . . . 258
23.4 Root test . . . 261

24 Power Series . . . 267
24.1 Convergence of a power series . . . 267
24.2 Radius of convergence . . . 269
24.3 Alternating series . . . 271

24.4 Rearrangement and conditional convergence 274

25 Taylor Series 279
25.1 Taylor series 279
25.2 Applications and two other versions 285
25.3 Convergence order of Gauss–Legendre 290

Part V Appendix

26 Energy of Planet Orbits 295
26.1 Potential energy in space 296
26.2 Energy of circular orbits 296
26.3 Energy of elliptical orbits 298
26.4 Interstellar and solar system objects 301

27 Elliptic and Hyperbolic Orbits 303
27.1 Eccentricity and focus of an ellipse 303
27.2 Directrices and ellipses 304
27.3 Polar equations of an ellipse 307
27.4 Display solar system orbits 309
27.5 Orbit of ICBM #2 310

28 Calculus for Planet Orbits 315
28.1 Variable centered notation 315
28.2 Coordinate system 316
28.3 Motion in polar coordinates 316
28.4 Newton's law of motion and gravitation 317
28.5 Two-body problem 318
28.6 Kepler problem 319

Index 321